Target-Weighted Loss Supplementary Analysis

Error-distribution analysis across target bins, illustrated in Supplementary Fig. S2, further showed that the model trained with standard MSE exhibited a pronounced bias toward low-value regions, with error increasing substantially in higher-value bins. In contrast, the weighted-loss model produced a more stable error profile, reduced mean error in elevated G_d and G_l ranges, and tightened the interquartile spread in high-value bins. Qualitative examples, as illustrated in Figs. S4 and S5 also showed that weighted-loss training better preserved focal high-value regions and spatial heterogeneity, whereas standard MSE training produced smoother maps with more pronounced underestimation in elevated regions.

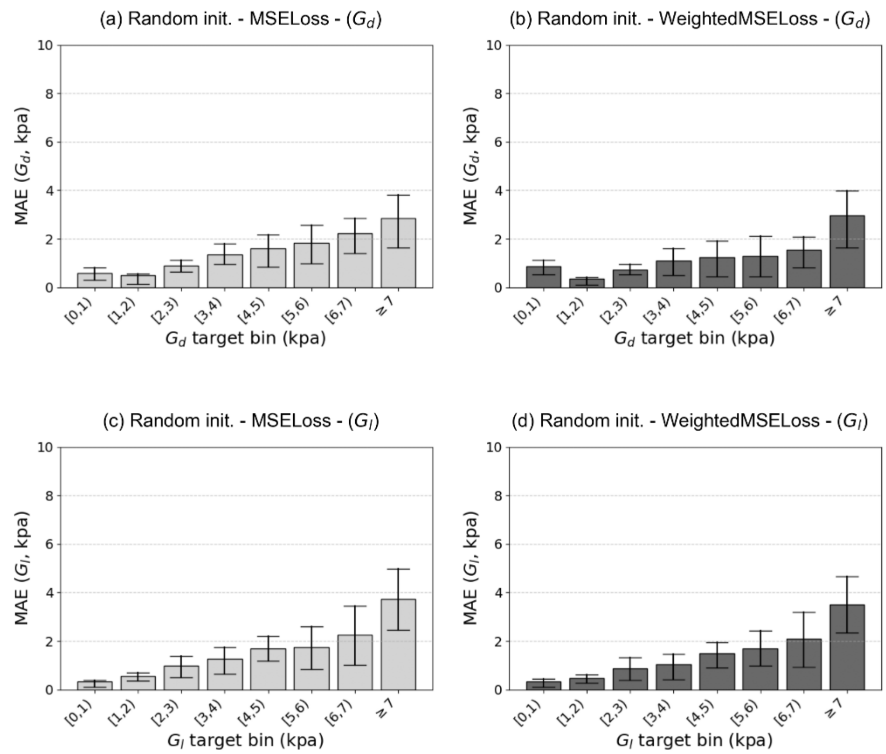

Fig. S2. Bin-wise mean absolute error (MAE) for storage modulus (G_d) and loss modulus (G_l) predictions on the test set for the loss-function comparison. (a) G_d error distribution for the model trained with standard MSE loss. (b) G_d error distribution for the model trained with standard MSE loss. (b) G_d error distribution for the model trained with the proposed Target-Mask-Weighted MSE loss. (c) G_l error distribution for the model trained with standard MSE loss. (d) G_l error distribution for the model trained with the proposed Target-Mask-Weighted MSE loss. The weighted-loss model shows lower error in underrepresented higher-value bins and a more stable error profile across the target range.

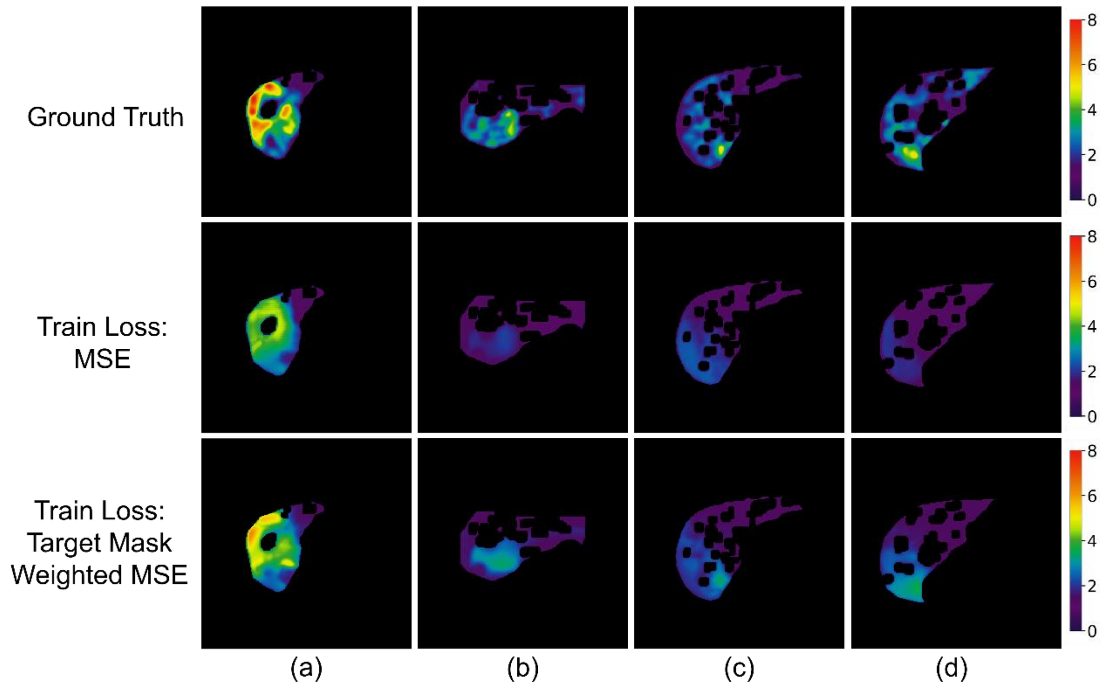

Fig. S3.Qualitative comparison for the loss-function experiment. Top row: ground-truth loss modulus (G_d) maps obtained from 3D-MRE. Middle row: predictions from the U-Net trained with standard MSE loss. Bottom row: predictions from the U-Net trained with the proposed Target-Mask-Weighted MSE loss. The weighted-loss model better preserves elevated-value regions and spatial heterogeneity in the liver parenchyma.

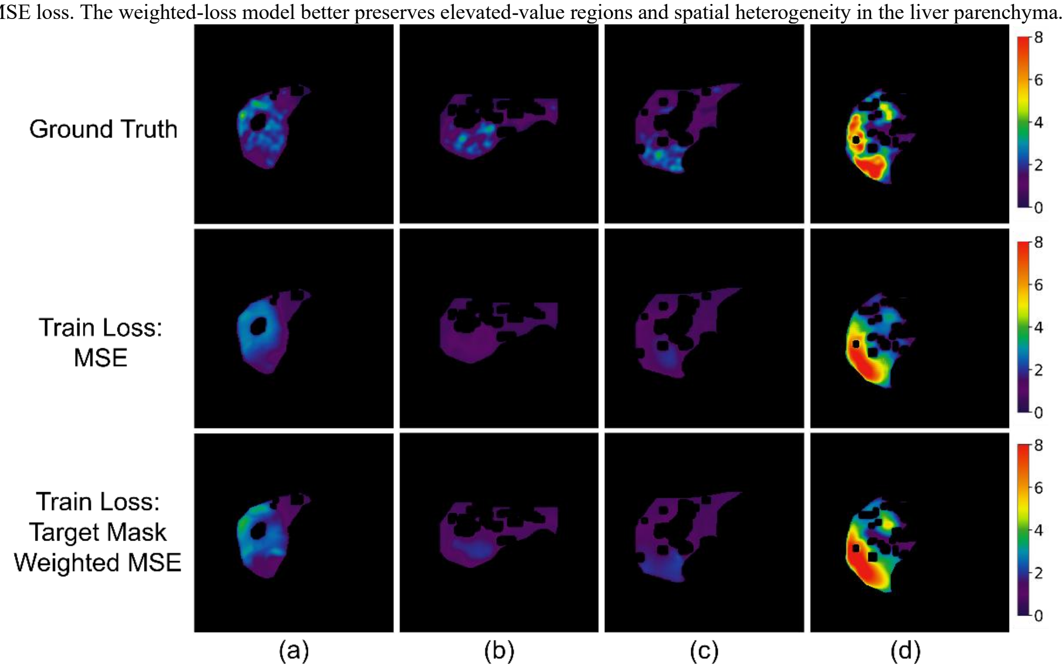

Fig. S4. Qualitative comparison of loss functions on G_l estimation. Top row: ground-truth loss modulus (G_l) maps obtained from 3D-MRE. Middle row: predictions from the U-Net trained with standard MSE loss, showing a tendency to underestimate high loss modulus values. Bottom row: predictions from the U-Net trained with the proposed Target-Mask-Weighted MSE loss, demonstrating improved retention of loss modulus heterogeneity and better definition of high-value regions.